



Magnum Technology Center
A WOM group company



Specialized in providing
integrated solutions
with advanced delivery to
the oilfield industry

Vision



To become a major supplier to the Well Test, Underbalanced Drilling and Managed Pressure Drilling sectors of the global oil and gas industry by providing an innovative range of individual and packaged products designed and manufactured to the latest industry standards from stock or with an advanced delivery and therefore enhance the ability of our customers to bring their services to the market on time and at a competitive price.



WORLDWIDE OIL

Mission



To build long term relationships with our customers by providing them with Well Test, Underbalanced Drilling and Managed Pressure Drilling equipment packages with innovative designs using the latest technology delivered from stock or with an advanced delivery backed up by a superior level of after sales support from our team of service industry experienced professionals and our extensive manufacturing and storage facility.



Seamless Integration

Magnum Technology Center designs and manufactures complete equipment packages for Well Testing, Underbalanced Drilling and Managed Pressure Drilling services. We're meticulous about the little details that make our packaged solutions so seamless. Everything has been designed to work together perfectly whilst giving operators the flexibility to adapt. With a huge depth of hands on experience from across the globe our products are designed with the realities of real life operations in mind. Fully certified to industry and latest ISO manufacturing standards and backed up by our extensive operational experience and customer support, MTC packaged solutions let you get on with the job in hand safe in the knowledge that we've done ours.

**WELL TESTING
& PRODUCTION**

This is what we do

**MANAGED
PRESSURE
DRILLING**



UNDERBALANCED DRILLING



The Magnum Technology Center
is a company specialized in providing integrated solutions
with advanced delivery to the oilfield industry.

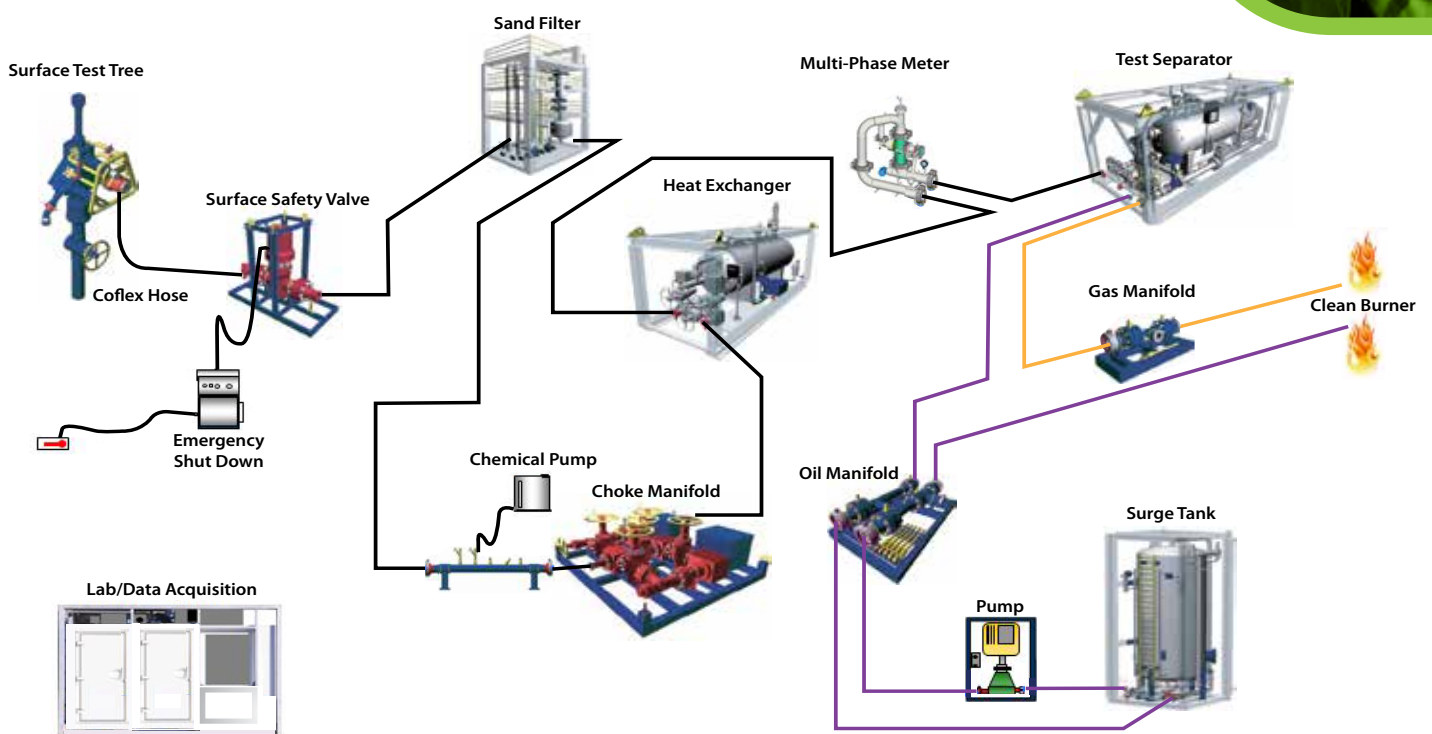
MAGNUM TECHNOLOGY CENTER

WELL TESTING

We provide the
complete package
for well testing companies



WELL TESTING GENERIC LAYOUT





Trailer Mounted Packages

Twin Trailer Concept

Cost Conscious Packages

From choke to burner head

Offshore/Onshore Packages

From flowhead to burner boom or burner head

Standard packages have been engineered and manufactured by MTC to satisfy the high standards demanded by the oil & gas industry and are delivered in a ready to use condition with all the required instrumentation, ancillaries and inter connections. Our packages are very versatile and can be used in the vast majority of service scenarios.

Skid mounted and Trailer mounted Well testing equipment is used by the operating oil companies to obtain accurate well performance and fluid characteristic data from oil and gas wells in varying drilling and production scenarios which is used in the evaluation of the reservoir under review.

Equipment is manufactured as per codes below:

- API 6A, ASME VIII Div.1 U Stamp, ANSI B31.3, Nace MR01.75, DNV 2.7-1 and ISO CSC with options of DNV OS-E-101 and NORSOK when required.
- TPI inspection by DNV, BV or Lloyds is provided as standard.

MTC provide packaged Well testing equipment sets in the below listed formats to match the needs of the global Well test service providers.

- Trailer mounted package equipment.
- Cost conscious package equipment.
- Offshore/ Onshore full package equipment sets from the Flowhead to the Burner booms.

TWIN TRAILER WELL TEST PACKAGE

Primary Well Test Process equipment Trailer is fully rated to the Zone 1 electrical standard with the the below equipment installed.

- 3" 10K Psi 5 Valve Choke manifold with integral Data header and ESD system.
- 1440 Psi 42" x 10ft 3 Phase Separator.
- 150 Psi, 50 BBL Twin compartment Surge tank with Steam coil.
- Twin 5000 BPD, 100 Psi Centrifugal Transfer pumps, electrically driven.
- Set of integrated interconnecting pipework and Explosion proof internal and external lighting package.

Secondary Well Test Trailer is an Office, Workshop and Utility unit for Safe Area use with the below equipment installed.

- 20ft Office and Workshop container with DAQ capability.
- Workbench with Pipe and Chain vices for operations & maintenance.
- 135KVA Diesel engine sound attenuated Generator with distribution / changeover system and safe area internal and external lighting package.
- Twin motor Compressor unit with large tank.
- 250ft of rig up pipework fitted in underslung racks both sides of trailer.





Office Utility Trailer



Twin Trailer



Well Test Trailer

COST CONSCIOUS WELL TEST PACKAGE

Well test cost conscious package is designed and packaged by MTC in a cost effective way to serve the service companies with equipment which is mounted in 2.5R load tested skid and frame in a ready to go format for any land jobs along with the flexibility of packaging the equipment in a trailer mounted format if required. Equipment package consists of:

- 3inch 5Kpsi Surface Safety Valve
- 3inch 5Kpsi 4 valve Choke manifold with integral data header
- 1440psi 42inch x 10ft 3 phase Separator in 2.5R load tested skid and frame.
- 150psi 50Bbl Twin compartment Surge tank with steam coil in 2.5R load tested skid and frame.
- 5000BPD 150psi Centrifugal transfer pump with outlet check valve.
- 1440 Psi Oil and gas diverter manifolds.
- 15,000 BPD 4 head Burner.
- 250 Ft of pipework consisting of 3 inch 1502,3 & 4 inch 206.
- 20ft converted Office and Workshop container.
- Standard Ancillaries.



SSV



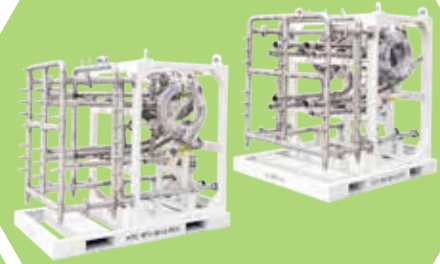
Choke Manifold



Separator



Oil-Manifold



Oil-Burner



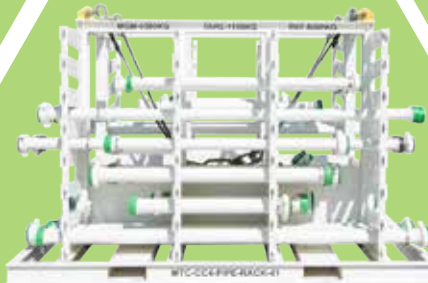
Transfer-Pump



Surge-Tank



Office Workshop-Container



Pipe Package



Gas-Manifold

OFFSHORE/ ONSHORE WELL TEST PACKAGE

Well test offshore /onshore package is designed by MTC to serve the service companies with equipment which is mounted in or can be transported in ISO-CSC and DNV skid and frame to be ready to go for work anywhere in the globe. Equipment package consists of,

- 3inch 10Kpsi Flowhead • 3inch 10Kpsi Surface Safety Valve
- 3inch 10Kpsi 5 valve Choke manifold with integral data header
- ESD system. • 3inch and 2 inch 10K Coflex Hoses
- 3inch 10Kpsi 4MMBtu Steam exchanger • 3 inch 5Kpsi 2MMBtu Indirect heater
- 1440psi 42inch x 10ft 3 phase Separator
- 150psi 80Bbl twin compartment Surge tank with steam coil
- 100 Bbl Twin compartment atmospheric tank
- 5000BPD 150psi Centrifugal transfer pumps with outlet check valve
- 1440 Psi Oil and gas diverter manifolds
- 15,000 BPD head burner • Burner Boom /King post / Burner head
- 950Ft of pipework consisting of 3 inch 1502, 3&4 inch 602,3&4 inch 206 in pipe racks
- DNV Accessories basket
- 12ft.Zone 1 A60 DNV 2.7-1 & 2.7-2 office container
- 10 ft DNV workshop / storage container
- 20ft converted Office and Workshop container
- Standard Ancillaries



Choke- Manifold



SSV



Flowhead



Surge Tank



Separator



Indirect-Heater



Pipe Package



Steam-Exchanger



Transfer Pump



A60 Zone1 Cabin



Gauge Tank



Burner Boom

WELL TEST ADDITIONAL EQUIPMENT

MTC also manufactures and stocks a large range of additional HPHT Well Test equipment items designed in house to complement the standard packaged equipment when required by our customers examples of which are listed below.

- 10K, 15K Psi and 20K Psi Twin Pot Sandfilter units in ISO-CSC & DNV Frames.
- 15K Flowheads and Surface safety valves in DNV baskets and frames.
- 15K Psi Steam Exchanger units in ISO-CSC & DNV Frames.
- 15K Psi Double Block & Bleed Choke manifold skids in flat 8 or Up & Over layouts in DNV skids.
- 10K Coflexip Flowline and Kill line hoses in DNV baskets.
- 1440 Psi 42" x 15ft DNV OS-E-101 and NORSOK Separators with Coriolis metering and 100 BBL, 250 Psi Twin compartment Surge tanks.
- 2000 Psi 42" x 10ft High Pressure Separators with Coriolis metering in DNV frames.
- Standalone Coriolis or Conventional Metering packages.
- 5,000 & 10,000 BPD Centrifugal Transfer pump packages in DNV frames.
- 3,000 & 4,000 BPD PCP Transfer pump packages in DNV frames.



15K Norsok Choke manifold



- 15,000BPD Onshore Burner heads with ignition systems.
- Operational spare parts and accessories packages.
- Wireless Data Acquisition packages.



2K Coriolis Separator



Norsok Twin Surge Tank



10K Twin Pot Sandfilter



Debris Catcher

WELL TEST ANCILLARIES



5DA Valve Control Panel



ESD-PTP-Panel



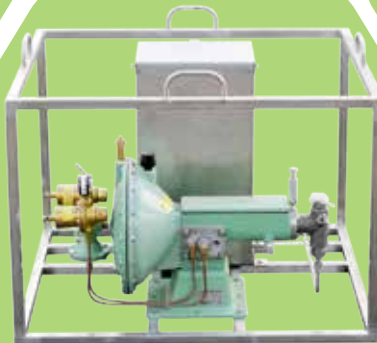
Pressure Recorder



Data Acquisition Package



Dead Weight Tester



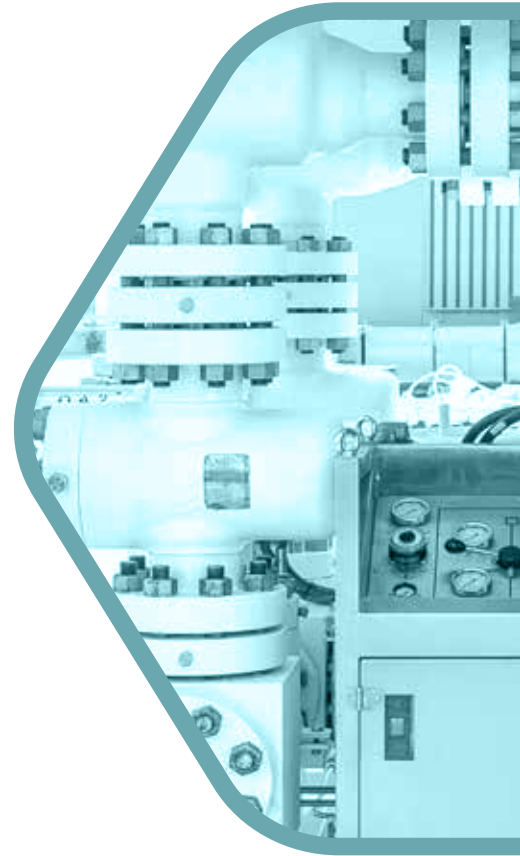
Chemical-Injection-Pump

Ancillary items provided from stock by MTC include but are not limited to the below.

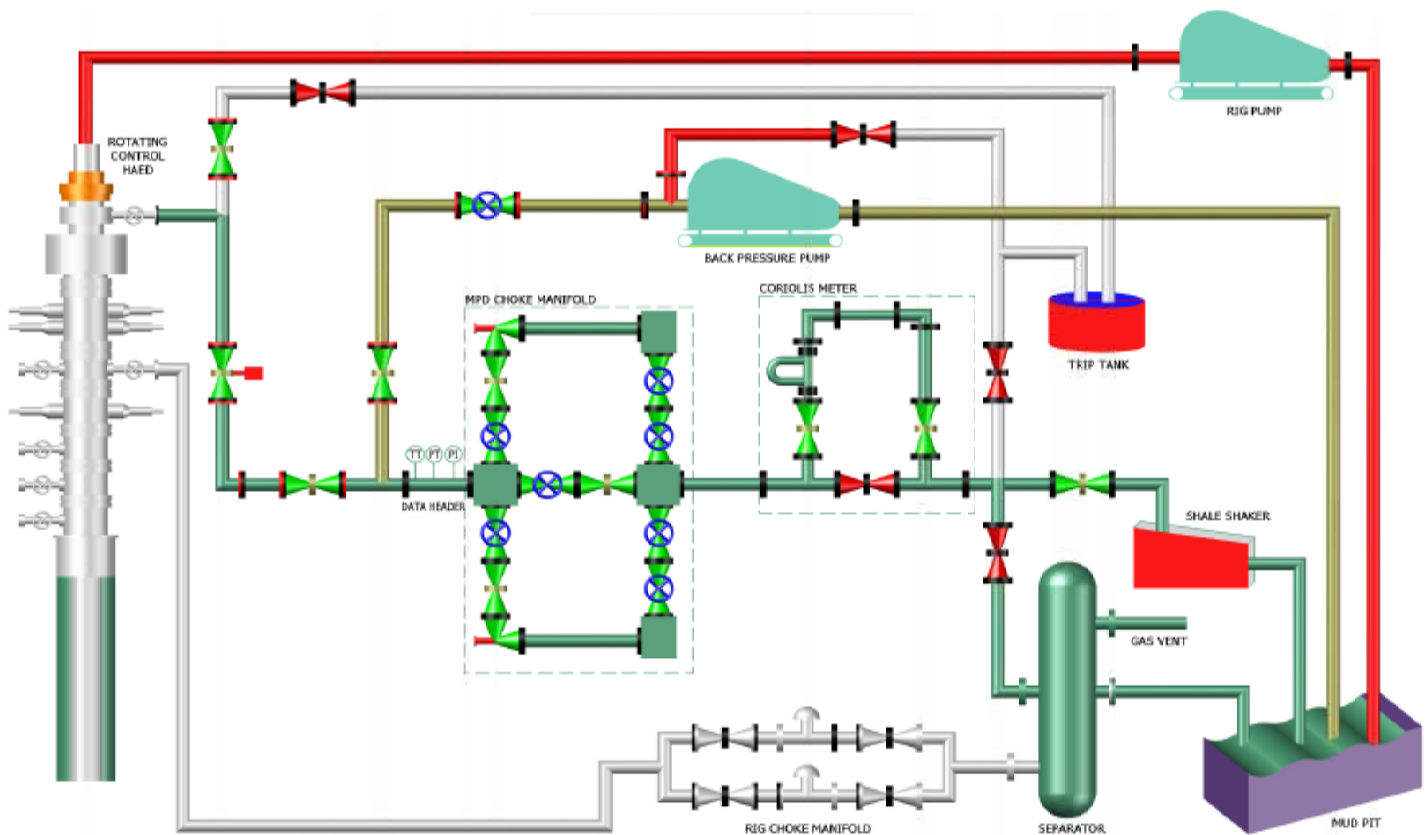
- 10K and 15K Psi Dead Weight Testers.
- 10K and 15K Psi Pressure Recorders.
- 10K and 15K Psi Chemical Injection units.
- 15K and 20K Psi Pressure test pump units.
- Single / Twin Emergency Shutdown Panels.
- Ranarex Gas gravity analysers.
- Electric / Air driven Centrifuges.
- Complete sets of Well Test lab accessories supplied in ready boxes.

MANAGED PRESSURE DRILLING

The perfect packages
for complete well control



MPD GENERIC LAYOUT





MPD Manifold Packages

MPD 5K Choke Manifold

MPD 2K Choke Manifold

MPD Buffer Manifold Packages

MPD 5K Buffer Manifold

MPD 3K Accumulator Rack

MPD 16DA Control Panel

MPD Coriolis Metering Skid Packages

MPD 2K Coriolis Meter

MPD #150RF Coriolis Meter

Our MPD choke manifolds are designed and manufactured by MTC to deliver accurate set point back pressure control via a user friendly HMI touch screen interface. The MPD equipment suite starts from the primary flow line to the MPD manifold and MGS through to the tie ins at the rig interfaces.

Standard package

- MPD 5 K Choke Manifold
- MPD 2 K Choke Manifold with Coriolis meter
- Mud Gas Separators
- Interconnecting pipe work in Pipe Racks
- Office Container pressurized (offshore)
- Office container Non- pressurized (Onshore)
- Workshop container
- Data Acquisition Systems

MANAGED PRESSURE DRILLING

Managed Pressure Drilling (MPD) equipment is used by the operating oil companies during drilling operations to complement the existing drilling package when drilling through difficult reservoir sections which require to be drilled balanced or slightly underbalanced to avoid kick/loss situations which can result in very expensive lost drilling time.

MTC manufactures both a high pressure API 5000psi gate valve type MPD manifold for offshore and onshore use and also a lower pressure ANSI #900 2000psi Ball valve type MPD manifold.

MTC also manufactures a range of skid mounted and trailer mounted Underbalanced Drilling (UBD) equipment which can include the integration of the above MPD equipment when required.

MTC also maintains a stock of mass flow Coriolis metering manifold skids and a range of high pressure 5Kpsi and low pressure 2Kpsi large bore piping packages on transportation racks for MPD or UBD applications.

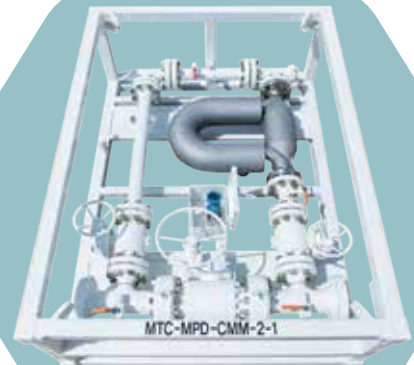
MTC can also provide additional equipment such as specifically designed mud gas separator (MGS) skids on a made to order basis.



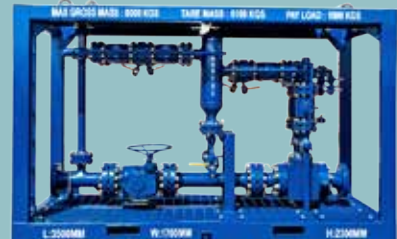
5K Manifold



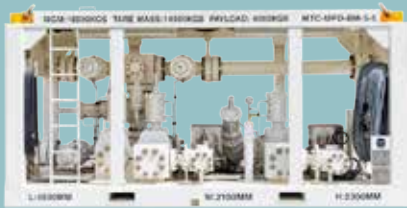
2K Manifold



2K Coriolis-Meter-Manifold



Sample-Catcher-Manifold



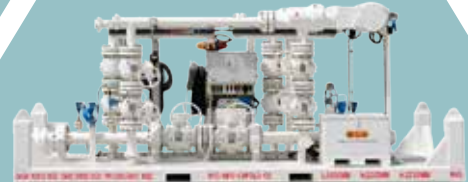
5K Buffer Manifold



Accumulator Rack



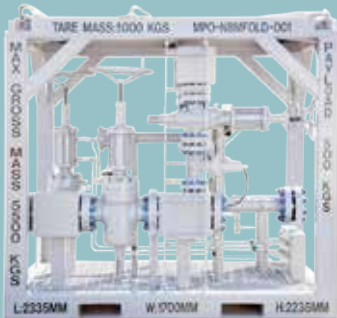
16 DA Control Panel & Remote HMI



2K Manifold



TCD Manifold

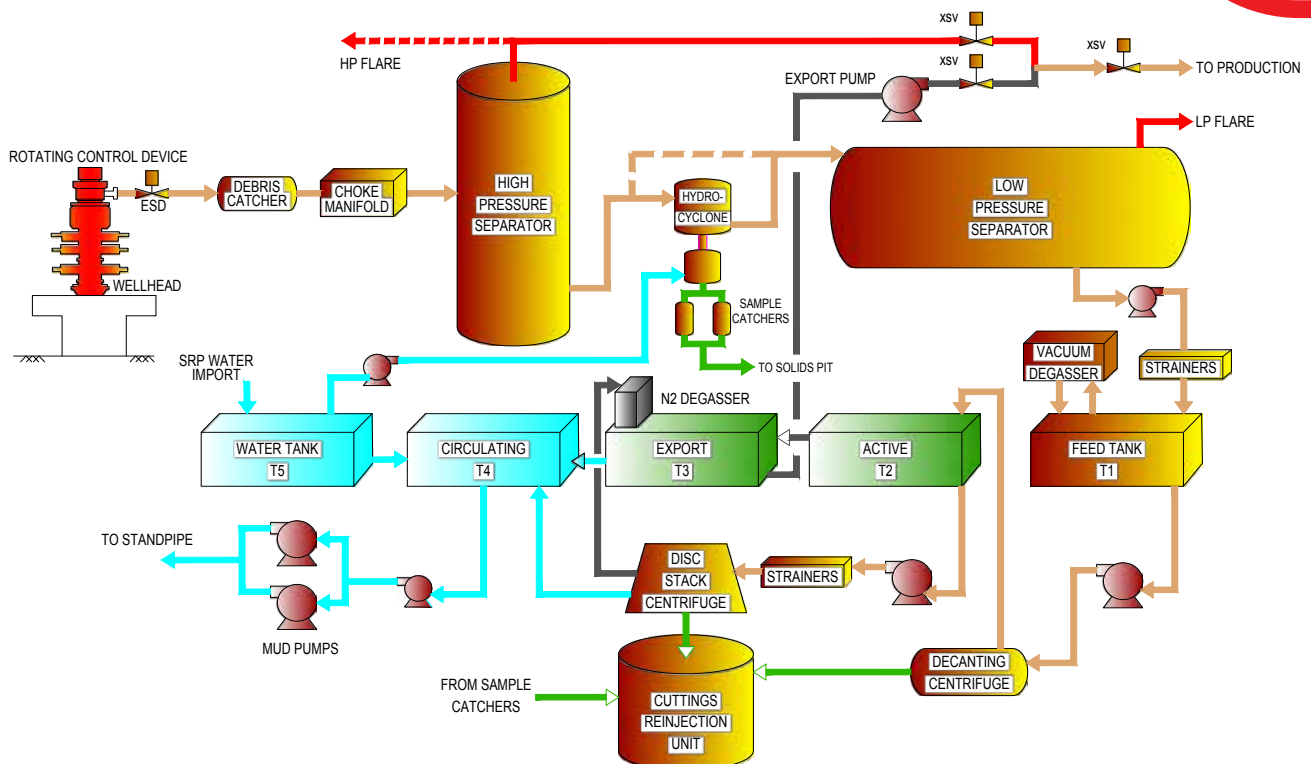


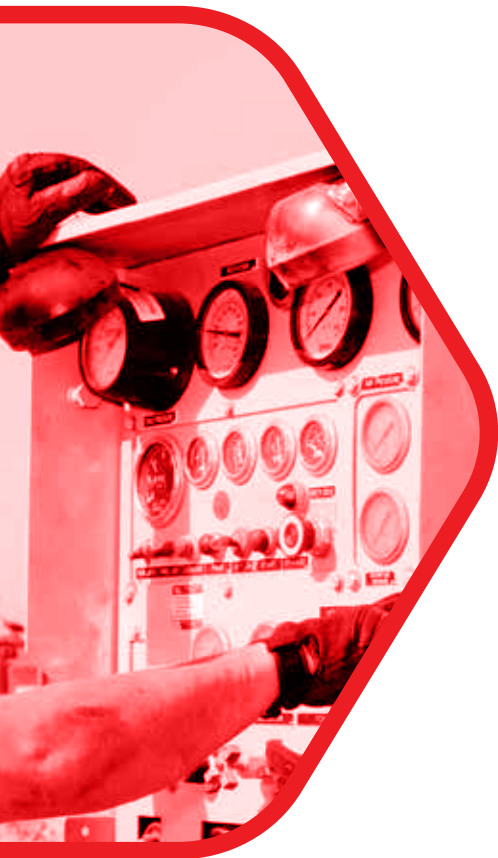
NSD Manifold

UNDERBALANCED DRILLING

The one stop shop for
integrated trailer
mounted well service
equipment package solutions

UBD GENERIC LAYOUT





UBD Integrated Trailer Mounted Process Packages

Well control and Sample catcher trailer

High pressure 3 Phase separator and Solids / Liquids Control trailer

Low pressure 1st Stage separator and Pump package trailer

Low pressure 2nd Stage separator and Control room trailer

Gas flare and Vent system Header manifold

UBD Integrated Skid Mounted Process Packages

Well control and Sample catcher skid

High pressure 3 phase separator skid

Solids / Liquids Control manifold skid

Low pressure 1st Stage Separator skid

Low pressure 2nd Stage Separator skid

Pump package and Controls skid

Office and Control room container

Gas flare and Vent system Header manifold skid

UBD Integrated Packages Ancillary Equipment

Oil / Condensate Re-injection Triplex Pump packages

Variable Frequency Drive containerised packages

Settling and Storage tank farm packages

Power generation and distribution packages

Offices, Workshops and Stores packages

Our UBD trailer and skid mounted equipment packages are designed and manufactured in house with the attention to detail that ensures the equipment can be deployed and rigged up rapidly to be ready to go online for operations in the minimum possible timeframe. The format and layout of the equipment on each trailer or skid is constantly under evaluation by our very experienced well services operations team during the design and manufacturing stages to make sure it is both operationally and maintenance friendly to optimize the troubleshooting and repairs of any real time issues that might develop during live operations as well as routine maintenance shutdowns.

UNDERBALANCED DRILLING

Underbalanced drilling equipment is utilized by the operating oil companies to complement the existing drilling rig package when drilling through difficult reservoir sections that require to be drilled underbalanced to avoid the damage that can be caused to sensitive reservoir formations by drilling fluids during conventional overbalanced drilling operations.

MTC manufactures both Trailer mounted and Skid mounted UBD packages for integration with conventional derrick or coil tubing type drilling rigs and can provide the primary UBD Choke control manifold(s) in both the 5Kpsi and 10Kpsi API 6A pressure range with various options of size and layout formats depending on the clients individual requirements. Choke manifold assemblies can be supplied from stock or with advanced delivery using our parent group companies range of industry standard high performance WOM 200M Gate valves with upstream sealing technology in both the manual handwheel and double acting hydraulic actuated options and including hydraulic actuated gate and seat type high performance erosion resistant positive shutoff adjustable chokes suitable for the UBD application.



Underbalanced Drilling Package



UBD Trailer Package



UBD Trailer Package

In addition MTC maintains a working stock of the main metering, flow control, level control, isolation and pumping components for the gas, liquids and solids process handling package downstream of the UBD manifold(s) to ensure we can meet our standard from stock or with advanced delivery advantage.



**MAGNUM
TECHNOLOGY
CENTER**

“Our newly expanded manufacturing complex is purpose built to deliver the highest quality products”



SEAMLESS INTEGRATION

Facility Capabilities

Manufacturing

Blasting & Painting

Refurbishment & Re-Certification

Pressure Testing - Hydro & Gas

Logistic Storage & Distribution

Extensive Spare Parts Inventory

Pressure Vessel, Process & Structural Design

Industry Standard Training Center of Excellence

MTC is strategically located in the Jebel Ali Free Zone to take full advantage of its duty free status, enhanced infrastructure and logistics support services. Our manufacturing facility currently allows us to design, engineer, manufacture and package our stock build range of well test and managed pressure drilling products.

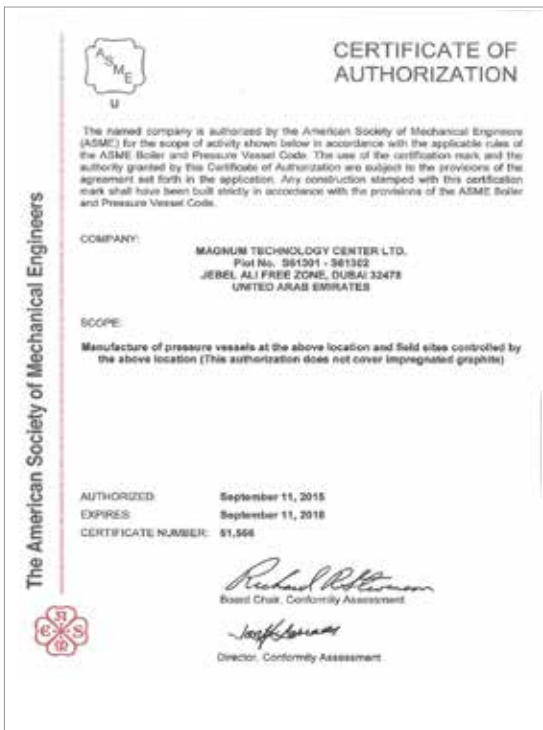
In line with continuing to meet the growing demand for our extensive range of packaged and standalone products we have completed the expansion of the complex to 36,000 sq meters (approx 10 acres) by adding a 12,000 sq meter fabrication and painting facility including ASME VIII Div 1 U Stamp & U2 capability and a 12,000 sq. meter logistics and storage yard.

As a further expansion to better support our current and future clients we now added a purpose built WOM Group Client Training Centre of Excellence.





A sale marks the beginning,
not the end of a relationship.





MTC also provides after sales support for OEM and non OEM equipment, with spare parts, maintenance and recertification support available from our Jebel Ali facility. Service engineers will be available for on site support on request.





A **WOM** group company

TRAINING CENTER OF EXCELLENCE



SEAMLESS INTEGRATION

Summary

Opened in November 2015 the WOM Training Centre of Excellence is available to our clients for specific training of your employees on equipment purchased from WOM/MTC and for general industry training in well testing, MPD, UBD and drilling operations related to WOM/MTC products. Our training department can also design, implement and manage your in house training and competency needs relieving you of time consuming tasks and costly overheads.

Equipment training will comprise both classroom

and practical sessions to enable the best possible learning experience.

Training can be tailored to meet client's specific needs and include competence assessments during practical sessions as required.

Well testing and MPD operations training will include maintenance and operational aspects of these services.

If you have your own instructors and/or equipment but nowhere suitable to train, our training centre is available for your offsite training requirements.

Our Facility



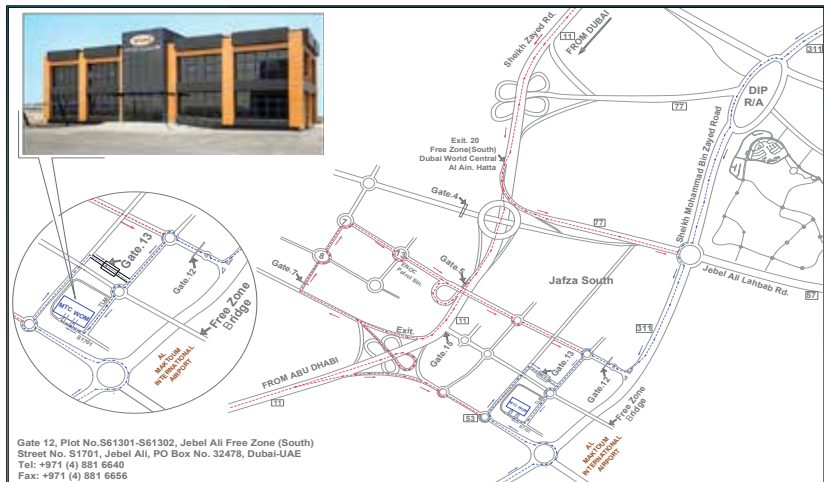
Three large classrooms each with a high level of natural light, air conditioned with adjustable seating plans to suit every need located on the upper floor of the building, each classroom has a ceiling mounted wireless projector. Two of the three classrooms can be combined by removal of a moveable, floor to ceiling, partition wall for large classes or conference requirements. A large air conditioned workshop, on the ground floor, for hands on practical training includes water, air and electrical utilities. A large shutter door allows items to be forklifted into the work area. The ground floor also has a large canteen and pantry suitable for outside catering supply. For training with larger equipment or packages our outside yard space will meet most requirements.

Where we are

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Classrooms



- Projector – wireless connection
- 16 seats at 8 desks in each classroom as standard, instructor's desk and presenting lectern. Flexible seating and desk arrangements to suit every need.
- Combine two classrooms with moveable, floor to ceiling, partition wall for large Training/conference requirements

Workshop

- 10m x 12m floor space
- Air conditioned
- Roller shutter linking outside for equipment to be forklifted into the work area, equipment up to height 3.0m and width 3.5m can be moved into the workshop for more comfort
- Large screen monitor
- Work benches and equipment cleaning station
- 16A and 32A Electrical Power outlets
- 120 psi air supply with McDonald connections



Services

- Industry standard training for well testing, MPD and drilling rig equipment
- Design and implementation of training and competency programs
- Management of client training and competency programs
- E-Learning design and delivery

Courses Offering

- Well testing & MPD
 - Non-technical Course
 - Basic Course for new employees
 - Planning, Operations, Reporting and Review
 - Well testing and MPD Equipment operation and maintenance
- WOM products theory and maintenance
 - Gate valves
 - Ball valves
 - Well test products
 - Subsea products

Conferencing

- Large configurable space for meetings and conferences



Magnum Technology Center

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